

# Harith Al-Safi

Full Stack Engineer

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## About me

An experienced Full Stack Engineer with **3+ years** of specializing in TypeScript, React, Java Spring Boot, and cloud technologies. Proven record in leading technical teams, and driving cost optimizations across **startup** to **enterprise**.

## Experience

### Full Stack Engineer

📅 Sep 2024 - Ongoing (12 Months)

BT Group

📍 Manchester, UK

- Delivered 5 **Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and Pulumi
- Mentored and examined 5+ apprentices on agile development with Java Spring Boot and **AWS Lambda**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing ECS cluster
- Re-architecture Spring Boot APIs with JWT, Azure AD, and **role-based access** achieving **enterprise grade security**
- Led **JavaScript to TypeScript** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Managed regional Graduate Network organizing and coordinating monthly tech talks for **30+ attendees**

### Frontend Developer

📅 Aug 2023 - Nov 2023 (4 Months)

Velz Travel

📍 Remote

- Directed the frontend of a travel web app for a startup built with **Redux Toolkit**, **Next.js** and Emotion CSS
- Developed user interface using **Material UI** following Figma design guidelines and business logic using TypeScript
- Achieved 93% test coverage through comprehensive unit testing, resulting in **70% reduction in production bugs**

### Software Engineer

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London, UK

- Invented a **custom scripting language with a compiler** for C# app increasing development productivity by 40%
- Managed a team of 10 interns to create a **business plan** and ended up placing 2<sup>nd</sup> amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure

### Full Stack Development Freelancer

📅 Dec 2021 - Mar 2022 (4 Months)

Paperound

📍 Leeds, UK

- Developed a scalable analytics dashboard in Next.js, with **MSAL** and **Auth0** for seamless, secure authentication.
- Developed **object detection** and tracking using OpenCV and Python for detecting planets in telescope images
- Developed **regression models** with Python, **Scikit-learn** and Matplotlib, and presented to business **stakeholders**
- Developed **Serverless REST API** with C# and Azure for **IoT Sensors** and analytics using SQL Triggers and Binding
- Engineered a **desktop QT** application using **C++** and **Cmake** for video analysis and processing

### IT administrator

📅 Jun 2020 - Mar 2021 (10 Months)

Mesopotamia Group

📍 Amman, Jordan

- Architected **scalable file server infrastructure** using Apache, Nextcloud, and Ubuntu in production environment
- Automated system administration workflows using **Python**, **Bash scripting**, and network security protocols
- Developed **high-availability load balancer with C/MPI** achieving 34% improved system uptime
- Built **predictive analytics pipeline using Python** for proactive cybersecurity threat detection
- Designed cloud networking solutions implementing TCP/IP, routing protocols, and infrastructure optimization

## Education

### BEng (Honours) in Electronics and Computer

📅 Sept 2019 - July 2024

University of Leeds (UoL)

✅ First-Class Honours (4.0/4.0 GPA)

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

## Skills and Languages

Technical:

TypeScript

Java

Python

C#

C \ C++

Bash

React.js

Next.js

Express.js

Redux

Tailwind CSS

Stripe

Auth0

Clerk

Spring Boot

Kafka

Conductor

.NET

Docker

Jira

Git

Matplotlib

TensorFlow

Numpy

SciPy

Pandas

NLP

MongoDB

PostgreSQL

Redis

SQLite

Jenkins

Firebase (Auth, Firestore)

AWS (RDS, Lambda, ECS, EC2, S3)

Azure (AD, MSAL, Serverless)

Pulumi

Soft:

Management

Leadership

Motivated

Deterministic

Communication

Problem solver

Languages:

Arabic (Full Professional)

English (Full Professional)

Spanish (Elementary)

## Publications

Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development

📅 2025

MPDI Sensors - 10.3390/s25123809 - 1<sup>st</sup> Author

- Developed a full stack **Chatbot** web application with TypeScript, React.js, Next.js, **TailwindCSS** and **Reddit**
- Programmed language processing algorithms using **multi-modal large language models** to execute user messages
- Implemented a **Python Flask** REST Server with **SQLite** on the **Raspberry Pi** to communicate with the chatbot
- Engineered a **circuit** with sensors, camera, fans, LCD and buttons to execute user's commands from the chatbot

## Projects

CV Studio

📅 Oct 2024 - Ongoing

👤 Personal - Beta on cvstudio.ai

- Built **full-stack** resume-improvement web app, powered by large language model, serving **hundreds of active users**.
- Implemented **live resume editing** with ShadcnUI and Typst **WebAssembly** PDF compilation for real-time render.
- Integrated **Clerk authentication** and **Stripe subscription** payments for secure user access and monetization
- Designed **Firestore real-time** database with Zustand state and Zod validation for seamless data syncing

Green ML: Machine Learning for Green Bond Syndication

📅 May 2024 - Aug 2024

👤 Personal

- Engineered bond syndication **prediction model** using **XGBoost** and network analysis on 3K+ institutional holders
- Implemented **parallel processing** with Ray to extract network features from adjacency matrices with **70% speedup**
- Built machine learning pipeline achieving **0.95+ AUC-ROC** for predicting bond syndication collaboration patterns
- Processed **Bloomberg financial data** using **pandas**, **NetworkX** for multi-regional investment analysis

MCPP: High Performance C++ Library for Numerical Computation

📅 Apr 2021 - Oct 2022

👤 Personal

- Created a **cross-platform math library** for C++ used for matrix algebra, statistics and machine learning.
- Engineered with **CI/CD** using **Jenkins**, unit tests, Docker build, **Cmake** and Doxygen documentation
- Improved performance by 40% using **parallel computing** with **OpenMP**, **OpenCL** and **CUDA**

## Professional Courses

Data Analysis with Python

📅 Oct 2021

- Perform exploratory **data analysis** and analytical techniques to datasets using Pandas, Numpy and Scipy
- Build and evaluate **regression models** using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python

📅 Sept 2021

- Analyze data within a relational database on the cloud using **IBM DB2**, **SQL**, **Python**
- Construct **SQL CRUD** operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development

📅 Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, **Numpy**, and Beautiful Soup
- Access web data using APIs and **web scraping** from Python in Jupyter Notebooks.